Home / Grade 6 / Geometry / Quarter 4

Nets of three-dimensional figures

Unfold the solid, and its surface area becomes an ordinary area on flat paper.

LESSON 182–184 | 3 × 40 MIN | MATEMATIKA 6, §36 | S7 8.4 3-D SHAPES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

What a net is	25 min
The nets of a cube	30 min
Nets of prisms, pyramids and cones	35 min
Surface area from a net	25 min
Homework	5 min

LEARNING OBJECTIVES

- Draw a net of a cube, a cuboid and a prism.
- Decide whether a given arrangement of squares folds into a cube.
- Name the parts of the net of a cylinder, a cone and a pyramid.
- Find a surface area by adding the parts of the net.

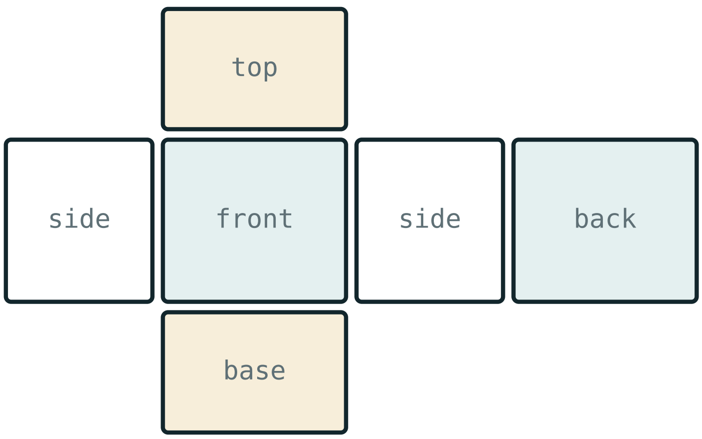
TEXTBOOK

National	pp. 522–531
Cambridge	Stage 7, pp. 90–96
Workbook	Exercise 8.6

01 Explanation

What a net is

A net is the solid cut along some of its edges and laid flat, with every face still joined to at least one neighbour. Fold it up and the original solid comes back.



surface area = the area of the whole net

The net of a cuboid — six rectangles in three equal pairs

SOLID	ITS NET
cube	six equal squares
cuboid	six rectangles, in three equal pairs
triangular prism	two triangles and three rectangles
square pyramid	one square and four triangles
cylinder	two circles and one rectangle
cone	one circle and one sector

THE AREA OF THE NET IS THE SURFACE AREA OF THE SOLID

Nothing is lost or added by folding. Every surface-area question in this course could be answered by drawing the net and adding its pieces.

The nets of a cube

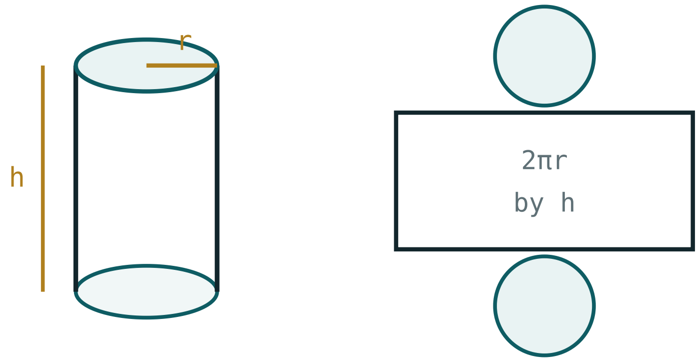
Six squares can be arranged in many ways, but only some of them fold into a cube.
There are exactly 11 genuinely different ones.

ARRANGEMENT	FOLDS INTO A CUBE?	WHY
a row of four with one square each side	yes	the classic cross
a row of three, two above, one below	yes	one of the eleven
a row of six	no	it wraps round and two faces land on the same place
a 2 by 3 rectangle	no	there is no square left for the top
four in a square with two attached	sometimes	it depends where the two are

THE TEST: WHICH FACES END UP OPPOSITE?

In a row of squares, the first and the fourth become opposite faces. If two squares of the net would have to become the *same* face, the arrangement fails. Cutting the shape out and folding it is the honest check — and the one worth doing.

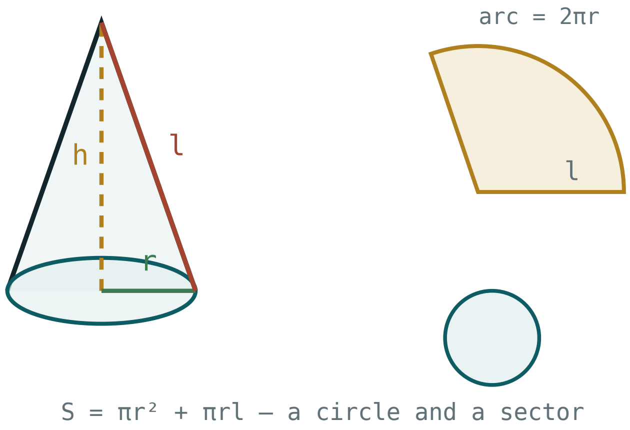
Nets of prisms, pyramids and cones



$S = 2\pi r^2 + 2\pi rh$ — two circles and one rectangle

Two circles and a rectangle whose width is the circumference

SOLID	PIECES OF THE NET	THE MEASUREMENT THAT SURPRISES
prism on an n-sided base	2 ends and n rectangles	each rectangle is as long as the prism
cylinder	2 circles and 1 rectangle	the rectangle's width is $2\pi r$
pyramid on an n-sided base	1 base and n triangles	the triangle's height is the slant height, not the solid's
cone	1 circle and 1 sector	the sector's radius is the slant height



The net of a cone — a base circle and a sector

THE SLANT HEIGHT IS NOT THE HEIGHT

In a pyramid the triangle leans outwards, so its own height is longer than the height of the solid. Using the wrong one is the single commonest error in net questions.

Surface area from a net

SOLID	THE PIECES	SURFACE AREA
cube, edge 4 cm	6 squares of 16	$96\ cm^2$
cuboid $5 \times 4 \times 3$ cm	$2(20 + 12 + 15)$	$94\ cm^2$
triangular prism, 3-4-5 ends, 10 cm long	$2 \cdot 6 + 30 + 40 + 50$	$132\ cm^2$
cylinder, $r = 5$, $h = 10$	$2 \cdot 78.5 + 314$	$471\ cm^2$
square pyramid, base 6, slant 5	$36 + 4 \cdot 15$	$96\ cm^2$
cone, $r = 3$, slant 5	$28.3 + 47.1$	$75.4\ cm^2$

LIST THE PIECES BEFORE ADDING ANYTHING

Write the shapes down — “two triangles of 6, three rectangles of $30, 40, 50$” — and the addition can hardly go wrong. Most lost marks here are a forgotten face, not bad arithmetic.

02 Worked examples

EXAMPLE 1 **Find the surface area of a $5 \times 4 \times 3$ cm cuboid from its net.**

1. The net has three pairs: 20, 12 and 15 cm².
2. $2(20 + 12 + 15)$.
3. $= 94$ cm².
   The same answer as the formula ✓

ANSWER

94 cm²

EXAMPLE 2 **A triangular prism has 3-4-5 cm ends and is 10 cm long. Find its surface area.**

1. Two triangles of $\frac{1}{2} \cdot 3 \cdot 4 = 6$ cm².
2. Three rectangles: $3 \cdot 10 = 30$, $4 \cdot 10 = 40$, $5 \cdot 10 = 50$.
3. $12 + 120 = 132$ cm².
   Five pieces, all listed ✓

ANSWER

132 cm²

EXAMPLE 3

A cylinder has radius 5 cm and height 10 cm. Find its surface area from the net, taking $\pi = 3.14$.

1. Two circles of 78.5 cm².

2. A rectangle 31.4 cm by 10 cm = 314 cm².

   Its width is the circumference.

3. $157 + 314 = 471$ cm².

ANSWER

471 cm²

03 Interactive model

Give out six paper squares taped in a row and ask the class to make a cube; the failure is more instructive than any diagram of the eleven nets.

Folding and unfolding

A net of a cube has:

- *4* squares
- *6* squares
- *8* squares
- *11* squares

The number of genuinely different nets of a cube is:

- *6*
- *8*
- *11*
- *24*

A row of six squares folds into a cube:

- yes
- no
- sometimes
- only if bent

The net of a cylinder is:

- two circles and a rectangle
- a circle and a sector
- six rectangles
- two triangles

The rectangle in that net has width:

- r
- $2r$
- πr
- $2\pi r$

The net of a cone is:

- a circle and a rectangle
- a circle and a sector
- two sectors
- a triangle

The triangles of a pyramid’s net use:

- the height of the solid
- the slant height
- the base only
- the volume

The area of a net equals:

- the volume
- the surface area
- half the surface area
- nothing useful

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Net	Yoyilma	Развёртка
To fold	Buklamoq	Складывать
Opposite faces	Qarama-qarshi yoqlar	Противоположные грани
Tab (flap)	Yopishtirish qismi	Клапан
Slant height	Yon balandlik (apofema)	Апофема
Sector	Sektor	Сектор
Circumference	Aylana uzunligi	Длина окружности
Surface area	Sirt yuzasi	Площадь поверхности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A net is the solid:

- cut in half
- cut along edges and laid flat
- drawn in perspective
- filled with cubes

A cuboid's net has:

- 3 rectangles
- 6 rectangles in 3 pairs
- 6 squares
- 4 rectangles

A prism on a hexagonal base has a net with:

- 6 rectangles and 2 hexagons
- 6 triangles
- 8 squares
- 2 circles

A 2 by 3 rectangle of squares folds into a cube:

- yes
- no
- sometimes
- only with tabs

The surface area of a cube of edge 4 cm from its net is:

- 16
- 64
- 96
- 24

The commonest error in pyramid nets is using:

- the base
- the height instead of the slant height
- too many triangles
- the wrong unit

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The faces in the net of a cube

   ANSWER 6

2. The number of different nets of a cube

   ANSWER 11

3. The net of a cylinder

   ANSWER Two circles and a rectangle

4. The net of a cone

   ANSWER A circle and a sector

5. The net of a square pyramid

   ANSWER A square and four triangles

6. The net of a triangular prism

   ANSWER Two triangles and three rectangles

7. The area of a net equals

   ANSWER The surface area of the solid

Medium · the standard question

7 PROBLEMS

1. The area of the net of a $5 \times 4 \times 3$ cm cuboid

   ANSWER 94 cm²

2. The area of the net of a cube of edge 4 cm

   ANSWER 96 cm²

3. The area of the net of a cube of edge 5 cm

   ANSWER 150 cm²

4. The width of the rectangle in a cylinder's net

   ANSWER $2\pi r$

5. That width for $r = 5$ cm, $\pi = 3.14$

   ANSWER 31.4 cm

6. The rectangles in the net of a hexagonal prism

   ANSWER 6

7. The triangles in the net of a pentagonal pyramid

   ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. The net of a triangular prism, 3-4-5 ends, 10 cm long

   ANSWER 132 cm²

2. The net of a cylinder, $r = 5$, $h = 10$

   ANSWER 471 cm²

3. The net of a square pyramid, base 6, slant 5

   ANSWER 96 cm²

4. The net of a cone, $r = 3$, slant 5

   ANSWER 75.4 cm²

5. Why does a row of six squares not fold into a cube?

   ANSWER It wraps round and two faces meet in one place

6. The area of the rectangle in the net of a cylinder $r = 5$, $h = 10$

   ANSWER 314 cm²

7. Which squares of a cube net must not touch?

   ANSWER Those that will become opposite faces

07 Homework

Homework — 5 tasks

Draw the nets on squared paper, list every piece, then add.

1. Draw a net of a cube of edge 3 cm and find its area.
2. Draw a net of a $6 \times 4 \times 2$ cm cuboid and find its area.
3. A cylinder has radius 4 cm and height 9 cm. Sketch its net and give the dimensions of the rectangle.
4. Find the total surface area of that cylinder, taking $\pi = 3.14$.

5. Draw two different nets of a cube, and explain why a row of six squares is not one of them.